

Week 7 — Cost-Benefit Memo: Model Selection for Phase 3

To: Dr. L. De Freitas, Mercer General ED Board, Martina Griffith (Clinical IT Lead) **From:** Mithra Ramoutar, Clinical AI Research Trainee **Re:** Is a more sophisticated model worth the extra complexity for Phase 3?

Verdict

Recommend Gradient Boosting for Phase 3, on the strength of its recall gain on ESI 1 patients — the metric we established as clinically primary — on the condition that every prediction is paired with a SHAP explanation to meet interpretability and audit requirements; none of the other complex models tested (Random Forest, tuned or untuned; a small neural network) beat the Week 6 baseline on any axis that matters clinically.

Recap: dataset and method

We are still predicting ESI (1–5) from triage-time vitals and chief-complaint flags only, using the same 55,121-encounter Yale EMLC dataset and the same 80/20 stratified split ((random_state=42)) as Week 6, so every comparison below is apples-to-apples. (disposition), (previousdispo), and the row-index column remain excluded as leakage. This week we engineered three additional clinical red-flag features (tachypnoea, hypoxia, fever, and a combined red-flag count) and tested three more complex models against the Week 6 logistic regression baseline: a Random Forest, a Gradient Boosting model, and a small neural network (MLP).

Benchmark table

Model	Accuracy	Macro F1	Recall ESI 1	Train time	Infer time/pred	Explainability
Baseline (LogReg)	0.667	0.492	0.250	2.8 s	0.003 ms	High
Random Forest	0.639	0.389	0.000	25.8 s	0.093 ms	Medium
Gradient Boosting	0.550	0.416	0.313	10.6 s	0.009 ms	Low
Small MLP	0.641	0.485	0.250	86.1 s	0.011 ms	Low

(Full table with precision/recall macro figures, the tuned-forest result, and the demographics-inclusion test are in [docs/week-7-benchmark.md](#).)

Three arguments for Gradient Boosting

1. **It is the only model that improves on the metric we said mattered most.** Week 6 established recall on ESI 1 as the primary metric, specifically because a missed critical patient is the costliest error in this system. Gradient Boosting catches 5 of 16 ESI 1 test patients versus the baseline's 4 of 16 — a real, if modest, improvement on exactly that number.
2. **Its inference cost is trivially small.** At 0.009 ms per prediction, it is well within any real-time triage workflow — Martina Griffith's "paid on every prediction, forever" cost is a non-issue here regardless of which model wins.
3. **Its macro-recall is also the best of the four (0.547),** meaning it is more willing to flag urgency broadly, not just for ESI 1 specifically — useful if the ED Board's real concern is under-triage in general, not just the rarest class.

Three arguments against Gradient Boosting

1. **It is the least accurate and least precise model tested.** Accuracy (0.550) and precision (0.410) are both the worst of the four — it generates meaningfully more false alarms than the baseline, which has its own operational cost (alarm fatigue, wasted resources on over-triaged patients).
2. **It is the hardest to explain.** Per the Week 7 tutorial's own interpretability ranking, gradient boosting requires SHAP to explain even a single prediction — Dr. Reyes cannot trace a decision by hand the way he could with the baseline's weights or a single tree. Adopting it means adopting SHAP as a permanent operational dependency, not a one-time analysis step.
3. **The absolute numbers behind this recommendation are small.** Only 16 ESI 1 patients exist in the test set; the entire "improvement" is one additional patient caught (5 vs. 4). A result built on a difference of one patient is not yet a result we should treat as settled — it needs to hold up under cross-validation and, eventually, local Mercer data, before it justifies a production decision.

Risks and unknowns

- **Sample size.** With only 77 ESI 1 encounters in the entire 55,121-row dataset, every ESI 1 recall figure in this memo is built on single-digit counts of test patients. These numbers should be treated as directional, not final, until validated with cross-validation and (eventually) a larger or locally-collected sample.

- **Demographics do not help.** We tested adding one-hot encoded `ethnicity` and `race` to the Random Forest and found no meaningful change in macro F1 (0.387 vs. 0.389 without them) — meaning there is no accuracy case for including fairness-sensitive columns in any model going forward, and we recommend they stay excluded regardless of which model is chosen.
- **Compute environment for these figures.** Random Forest and the tuned-forest search were run with a reduced tree count and search budget for this memo's benchmark run, due to the single-core environment used to produce it; a full re-run with the assignment's recommended settings (300 trees, 8-iteration/3-fold search) is planned before Phase 3 sign-off, though the gap between the baseline and the other models is wide enough that this is unlikely to change the recommendation.
- **No local validation yet.** As flagged since Week 5, every figure here is a US academic-hospital benchmark, not a guarantee of Mercer General performance.

Recommendation

Carry Gradient Boosting into Phase 3 piloting, paired with SHAP explanations at the point of each prediction, specifically because it is the only model tested that improves recall on the class we have already told the ED Board is our top clinical priority. This is not "the best model" by most axes — it is the most defensible trade given what we are optimising for. The baseline remains a credible fallback if SHAP integration proves too costly for Martina Griffith's IT Governance requirements; it should not be discarded.